

WEC2026 Goal-Scoring Prediction: Data, Preprocessing, and Modeling Report

Automated experiment report

April 26, 2026

1 Executive Summary

The task is a rare-event binary classification problem: predicting whether a player scores after a match checkpoint. The modeling table contains 3,486 checkpoint observations, 203 positives (5.82%). The best grouped cross-validated model in this run was **catboost** on **all_features** features with ROC-AUC 0.640, PR-AUC 0.134, and balanced accuracy 0.604. AutoGluon was not available in this run.

2 Data Audit

The checkpoint file provides player state every 15 minutes, including position, home status, formation, substitution times, physical load, shots, and the target **scored_after**. Four event files were joined using **player_appearance_id** and absolute match time: passes, high-speed runs/sprints, shots, and actions under defensive pressure. The target is highly imbalanced; therefore ROC-AUC, PR-AUC, balanced accuracy, Brier score, and log loss are more informative than raw accuracy.

3 Preprocessing

All event-derived features were computed with temporal filters:

$$\text{minute_in} \leq \text{event_abs_minute} \leq \text{checkpoint_abs_minute}.$$

The implementation converts period-relative minutes to absolute match minutes so second-half and extra-time checkpoints are handled correctly. Reference quantities such as expected-threat zone values, pressure turnover baselines, and positional baselines were fitted inside each cross-validation training fold before being applied to validation checkpoints. Categorical variables were one-hot encoded from training folds only, and numerical missing values were median-imputed from training folds only.

4 Feature Families

The all-feature model uses signals from every available file: checkpoint physical and shot summaries, run-event intensity, pass volume and xT proxies, shot context, pressure resistance, and pressing-applied metrics. Recent-vs-cumulative ratios were included to test short-term intensity surges. Before fitting, 18 deterministic or near-deterministic aliases were excluded from every model feature

set, leaving 151 raw all-feature columns before one-hot encoding. The screen removes duplicated run summaries, duplicated shot counts or ratios, pressure expected-turnover count proxies, and the raw position category once role indicators are present. The retained representatives preserve the same football information while reducing collinearity and stabilising permutation importance.

5 Model Selection and HPO

The benchmark used fixture-grouped stratified cross-validation to prevent match-level leakage. Models attempted in this practical run included logistic regression, ExtraTrees, RandomForest, XGBoost, CatBoost, LightGBM, TabPFN, stacking, and optional AutoGluon. TabPFN probabilities were included as a base learner in the stacked ensemble whenever the package was available, rather than being evaluated only as a standalone benchmark. Skipped or unavailable models: None. XGBoost used Optuna HPO when the requested trial budget was nonzero; other model parameters were conservative regularized defaults appropriate for the small sample size.

6 Grouped Cross-Validated Results

Model	Feature set	ROC-AUC	PR-AUC	Bal. Acc.	F1	Brier
catboost	all_features	0.640	0.134	0.604	0.175	0.158
random_forest	all_features	0.687	0.115	0.500	0.000	0.058
tabpfn	all_features	0.644	0.111	0.500	0.000	0.055
extra_trees	all_features	0.674	0.109	0.500	0.000	0.062
lightgbm	all_features	0.651	0.109	0.500	0.000	0.063
xgboost	base_plus_passing	0.642	0.105	0.499	0.000	0.066
xgboost	base_plus_pressure	0.634	0.105	0.499	0.000	0.066
xgboost	base_checkpoint	0.641	0.104	0.500	0.000	0.066
xgboost	cumulative_only	0.637	0.100	0.501	0.018	0.070
stacked	all_features	0.581	0.097	0.583	0.140	0.055
xgboost	all_features	0.629	0.096	0.502	0.010	0.066
xgboost	recent_only	0.640	0.093	0.502	0.010	0.069
logistic	all_features	0.596	0.093	0.566	0.136	0.204
xgboost	sprints_shots	0.584	0.070	0.496	0.015	0.089
xgboost	external_context	0.538	0.063	0.523	0.113	0.231

7 AutoGluon Holdout Result

Model	Feature set	ROC-AUC	PR-AUC	Bal. Acc.	F1	Brier
None recorded.						

8 Ablation Results

Feature set	ROC-AUC	PR-AUC	Bal. Acc.	F1
base_plus_passing	0.642	0.105	0.499	0.000
base_plus_pressure	0.634	0.105	0.499	0.000
base_checkpoint	0.641	0.104	0.500	0.000
cumulative_only	0.637	0.100	0.501	0.018
all_features	0.629	0.096	0.502	0.010
recent_only	0.640	0.093	0.502	0.010
sprints_shots	0.584	0.070	0.496	0.015
external_context	0.538	0.063	0.523	0.113

9 Feature Importance

Feature	PR-AUC drop after permutation
cumul_pass_bottom_share	0.031
recent_distance_ratio	0.010
is_defender	0.008
last15_pass_bottom_share	0.007
cumul_pressure_top_share	0.007
last15_pressure_top_share	0.003
last15_run_top_share	0.003
cumul_pass_top_share	0.003
cumul_pressure_quality	0.003
last15_peak_speed	0.003
is_attacker	0.003
last5_pass_bottom_share	0.002

10 Research Questions

RQ1. Player behaviour during a match is predictive but noisy because goals are rare. In this run the best grouped-CV model reached ROC-AUC 0.640, PR-AUC 0.134, and balanced accuracy 0.604.

RQ2. The strongest determinants are identified by a combination of ablation deltas and permutation importance. In this run they were dominated by pass-zone shares, recent distance intensity, role indicators, pressure-location metrics, and pressure quality; the exact ranked variables are reported in the feature-importance table.

RQ3. The sprint-and-shot-only ablation should be compared against the all-feature model. Its result in this run was 0.070 PR-AUC, so it is not assumed sufficient unless it matches the all-feature score.

RQ4. Passing and pressure data are useful when their ablation improves over the base checkpoint model. The report tables show whether `base_plus_passing` and `base_plus_pressure` improve ROC-AUC/PR-AUC/balanced accuracy.

RQ5. The `recent_only` and `cumulative_only` ablations compare short-term and accumulated performance. The stronger family is the one with higher grouped-CV PR-AUC and ROC-AUC.

RQ6. Recent-vs-cumulative ratios and intensity-surge interactions directly test whether short-term intensity above a player’s own baseline increases scoring probability.

RQ7. External factors were modeled through home status, period, checkpoint time, extra time, formation, role indicators, and checkpoint-available substitution timing. Their influence is assessed by the `external_context` ablation and permutation importance.

11 Limitations

Only 203 positive examples are available, so fold-level variance is expected. Some event locations are coarse three-zone labels, which limits xT precision. The stacked model is evaluated with grouped OOF meta-predictions, but it should still be validated on a hidden competition set before final claims.